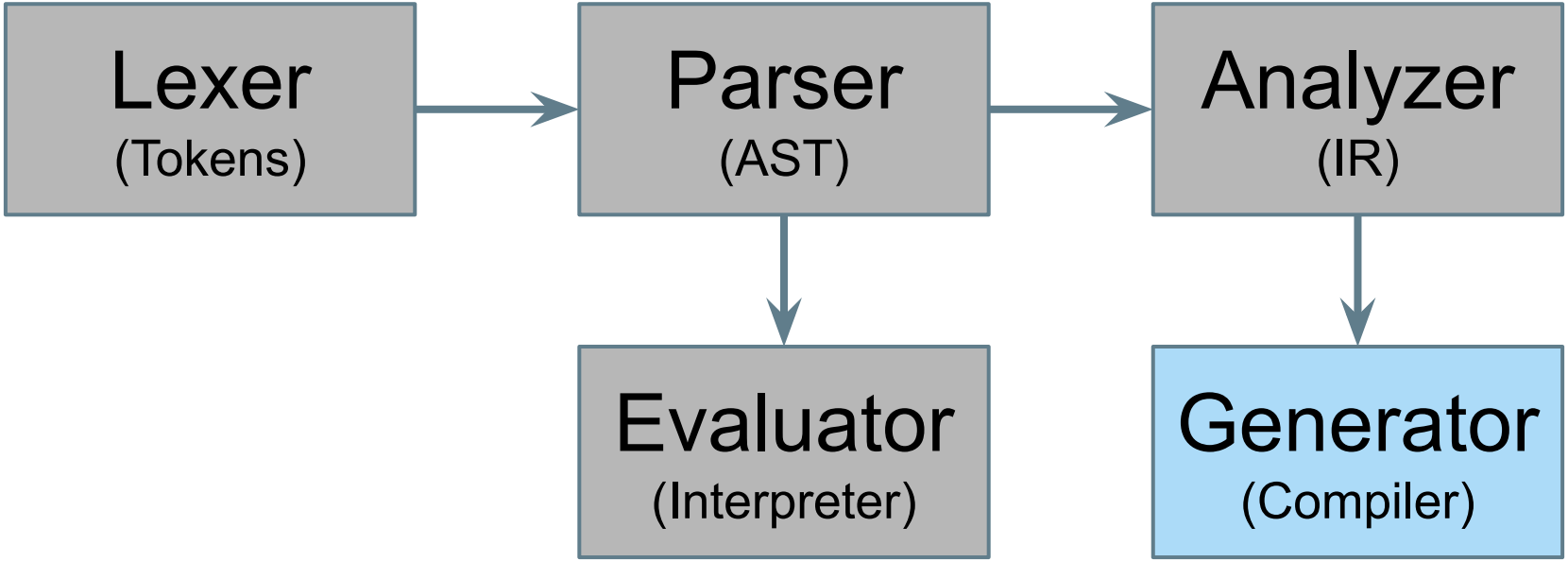
Module 6: Compilation

Compilation*

**or at least something close enough.*

The Big Idea (!)


Lexer
(Tokens)
Parser
(AST)
Analyzer
(IR)
Evaluator
(Interpreter)
Generator
(Compiler)

1.

Compiler Backends

Compiler Frontend/Backend

Frontend

The **frontend** of a compiler deals with the *source language* (input).

- Lexing/Parsing: Source -> AST
- Analyzer: AST -> IR stages
- Effectively, the frontend handles parsing/validation of the source.
  - information/understanding

Backend

The **backend** of a compiler deals with the *target language* (output).

- At *some point*, all programs are machine instructions for a specific architecture.
  - e.g. x86, ARM, MIPS.
- Everything above this is “higher level”; it’s an abstraction over the machine.
  - e.g. Java, C/C++, ASM, Python
- The backend handles the process for compiling to a specific target language.

Compilation Approaches

Interpreted

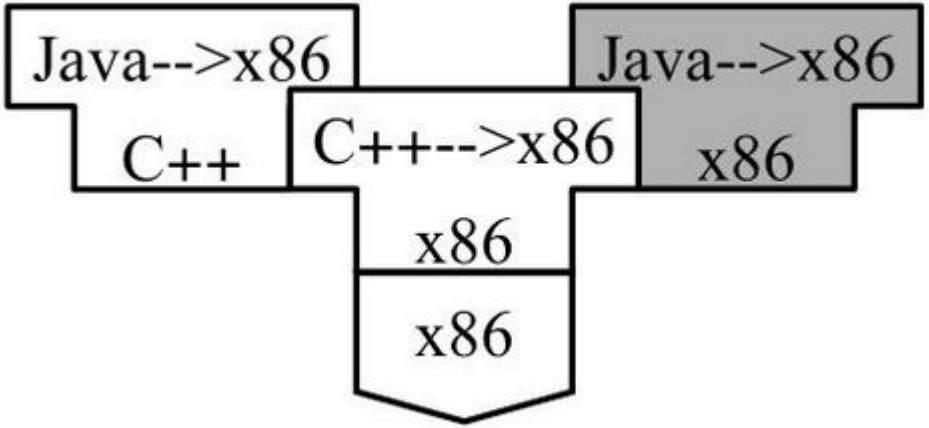
Converts an intermediate language into machine code instructions at runtime.

- “Tree Walk”: Easy, but slow.
  - Good for new/experiemental languages, or just functional.
- “Bytecode”: Harder, but faster.
  - Good for mature language, but deceptively complicated.
  - JVM is a Bytecode interepter.

Compiled

Produces a compiled output that can be run instead of the source code.

- Machine Code Compilation: Target a specific machine architecture.
- “Transpilation”: Leveraging an existing language as the target for compilation.
  - e.g. Lang -> C/C++ -> ASM -> x86
  - Effectively, this is all compilation now.
  - Limited by the abilities of the target language (e.g. memory management).



“T” Diagrams: Compiler Input, Output, & Implementation

A note on Compiler Optimizations

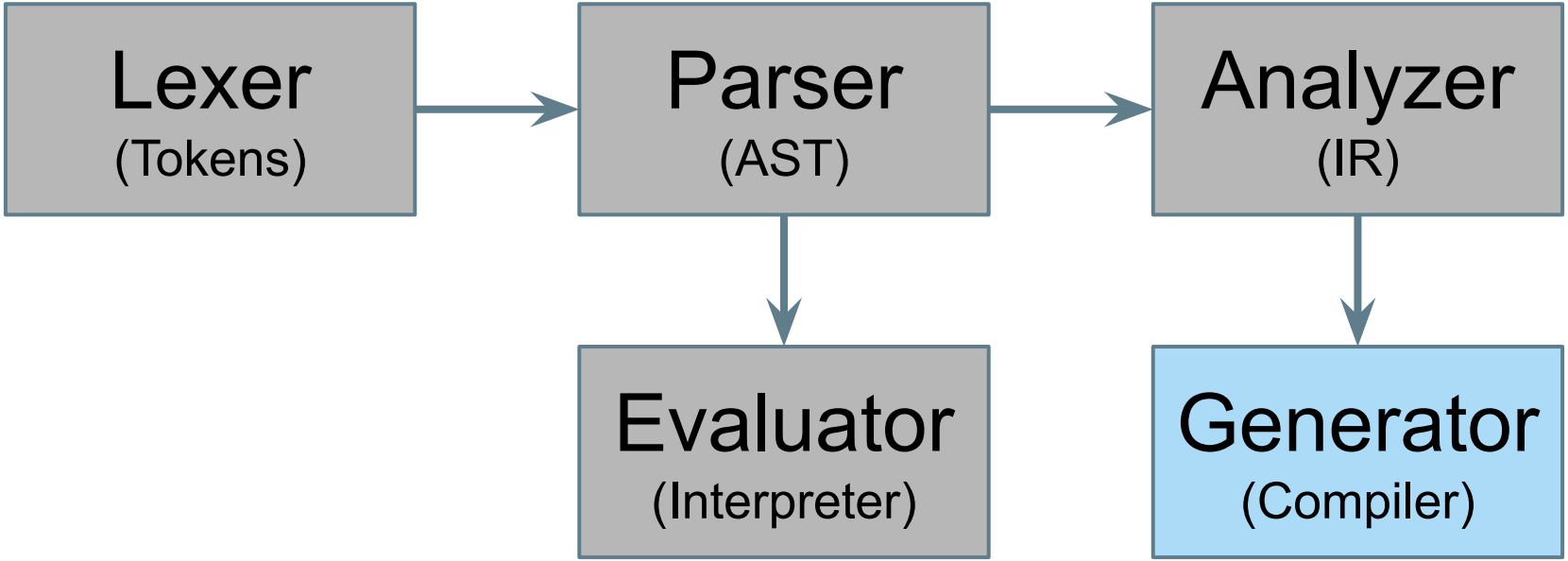
1. Really cool! But, not particularly helpful.
2. Requires more sophisticated architectures.

2.

Generator Introduction

Who needs a compiler backend when you have the compiler?

The Big Idea (!)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Compilation

Time for a Module 6 crash course in one lecture.

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